Lecture 2

A driven qubit: magnetic resonance

Objectives. *A proton in a strong magnetic field is a qubit with two energies. A second field, two thousand times weaker, can turn its spin over completely, provided that it rotates at the right frequency. This lecture explains how. The weak field changes in time, and so does the Hamiltonian. We solve the Schrödinger equation exactly in a frame that turns with the field, and find Rabi's formula for the probability that the spin has turned over. Two short pulses with a long wait between them then give fringes far narrower than the resonance.*

2.1 Turning a spin over

We write $|\uparrow\rangle = |0\rangle$ and $|\downarrow\rangle = |1\rangle$ for the states whose Bloch vectors are $\mathbf{e}_z$ and $-\mathbf{e}_z$, spin up and spin down along z. A spin in a strong magnetic field along the z-axis has, by Lecture 1, a Hamiltonian $H = \frac{\hbar\nu_0}{2}\sigma_z$ with $\nu_0 > 0$. The Bloch vector precesses about $\mathbf{e}_z$ at the angular frequency ν_0. The states $|\uparrow\rangle$ and $|\downarrow\rangle$ are its stationary states. How can we take a spin from $|\uparrow\rangle$ to $|\downarrow\rangle$?

By Theorem 1.12, a constant Hamiltonian keeps $\mathbf{n}\cdot\mathbf{r}$ constant. It can turn $\mathbf{e}_z$ into $-\mathbf{e}_z$ only if $\mathbf{n}\cdot\mathbf{e}_z = -\mathbf{n}\cdot\mathbf{e}_z$, that is, only if its axis $\mathbf{n}$ is perpendicular to $\mathbf{e}_z$. Add a weak static field along x. The Hamiltonian becomes

$$H = \frac{\hbar}{2}\big(\nu_0\sigma_z + \Omega\sigma_x\big), \qquad 0 < \Omega \ll \nu_0.$$

Its axis is the unit vector $\mathbf{n}$ in the direction of $(\Omega, 0, \nu_0)$, tilted from $\mathbf{e}_z$ by the small angle α with $\tan\alpha = \Omega/\nu_0$. The precession keeps the component $\mathbf{n}\cdot\mathbf{r}$ constant. So the Bloch vector of a spin prepared up stays on the circle of latitude about $\mathbf{n}$ through $\mathbf{e}_z$. The lowest point of that circle has $r_z = \cos 2\alpha$. By Corollary 1.8 with $\mathbf{m} = \mathbf{e}_z$, the probability of reading $|\downarrow\rangle$ is $\frac{1}{2}(1 - r_z)$, so it never exceeds

$$\frac{1}{2}(1 - \cos 2\alpha) = \sin^2\alpha = \frac{\Omega^2}{\Omega^2 + \nu_0^2}.$$

For $\Omega/\nu_0 = 5\cdot 10^{-4}$ this is $2.5\cdot 10^{-7}$. A weak static field tilts the axis a little, and the spin barely moves.

A field that turns does better. The spin precesses about $\mathbf{e}_z$ at the rate ν_0. A transverse field that turns about $\mathbf{e}_z$ at the same rate keeps a fixed angle to the precessing spin, so its effect builds up instead of averaging out. The rest of the lecture computes how far.

Self-check 2.1

1. A static transverse field has $\Omega/\nu_0 = 0.1$. What is the largest probability of reading $|\downarrow\rangle$ for a spin prepared up?

2. Which states does $H = \frac{\hbar}{2}(\nu_0\sigma_z + \Omega\sigma_x)$ leave where they are?

2.2 A Hamiltonian that changes in time

The coils of the experiment make a transverse field of fixed strength that turns about $\mathbf{e}_z$ at the angular frequency $\nu > 0$. The Hamiltonian is

$$H(t) = \frac{\hbar\nu_0}{2}\,\sigma_z + \frac{\hbar\Omega}{2}\big(\cos(\nu t)\,\sigma_x + \sin(\nu t)\,\sigma_y\big), \qquad \nu_0 > 0,\ \Omega > 0. \tag{2.1}$$

It is the Hamiltonian of a *circular drive*. The number Ω is the *Rabi frequency*, the precession rate that the transverse field alone would cause. The field is set by the experimenter, and we treat it as a given function of time, not as a quantum system.

Postulate 1.10 holds with $H(t)$ in place of H:

$$\mathrm{i}\hbar\,\frac{\mathrm{d}\psi_t}{\mathrm{d}t} = H(t)\psi_t.$$

At every time $H(t)$ is self-adjoint. The proof of Lemma 1.11 differentiates $\langle\psi_t, \psi_t\rangle$ at one time and uses only that the Hamiltonian is self-adjoint at that time. So the norm $\|\psi_t\|$ is constant here too, and the total probability stays 1.

Lemma 1.11 also solved the equation, by the exponential of the constant matrix $-\mathrm{i}tH/\hbar$ (*Linear Algebra*, Prop. 6.21). That proposition needs a constant matrix, and $H(t)$ is not constant. We shall not need a theory of equations with variable coefficients. A change of variables removes the time dependence of (2.1) exactly.

Self-check 2.2

1. Write $H(t)$ at $t = \pi/(2\nu)$ in terms of the Pauli matrices.
2. Why does $\|\psi_t\|$ stay constant although H changes in time?

2.3 The rotating frame

Let

$$R(t) = e^{-\mathrm{i}\nu t\sigma_z/2} = \begin{pmatrix} e^{-\mathrm{i}\nu t/2} & 0 \\ 0 & e^{\mathrm{i}\nu t/2} \end{pmatrix}.$$

By Lemma 1.11, $R(t)$ is the evolution under the Hamiltonian $\frac{\hbar\nu}{2}\sigma_z$ for the time t. So by Theorem 1.12 it turns the Bloch sphere about $\mathbf{e}_z$ through the angle νt:

$$\mathbf{r}\big(R(t)\chi\big) = \mathrm{Rot}_z(\nu t)\,\mathbf{r}(\chi) \qquad \text{for every unit vector } \chi, \tag{2.2}$$

where $\mathrm{Rot}_z(\beta)$ is the rotation of $\mathbb{R}^3$ about $\mathbf{e}_z$ through the angle β, anticlockwise seen from the tip of $\mathbf{e}_z$. Put $\Delta = \nu_0 - \nu$, the *detuning* of the drive, and

$$\tilde{H} = \frac{\hbar}{2}\big(\Delta\,\sigma_z + \Omega\,\sigma_x\big). \tag{2.3}$$

Lemma 2.1 (The rotating frame). *Let $H(t)$ be the Hamiltonian (2.1).*

1. $H(t) = R(t)\,\frac{\hbar}{2}\big(\nu_0\sigma_z + \Omega\sigma_x\big)\,R(t)^\dagger$.
2. *A differentiable ψ_t solves $\mathrm{i}\hbar\,\dot\psi_t = H(t)\psi_t$ if and only if $\tilde\psi_t = R(t)^\dagger\psi_t$ solves $\mathrm{i}\hbar\,\dot{\tilde\psi}_t = \tilde H\tilde\psi_t$.*

3. *For every unit vector ψ, the only solution with $\psi_0 = \psi$ is*

$$\psi_t = R(t)\, e^{-\mathrm{i}t\tilde{H}/\hbar}\, \psi. \tag{2.4}$$

Proof.

1. The matrix $R = R(t)$ is diagonal, so $R\sigma_z R^\dagger = \sigma_z$. Next,

$$R\,\sigma_x R^\dagger = \begin{pmatrix} 0 & e^{-\mathrm{i}\nu t} \\ e^{\mathrm{i}\nu t} & 0 \end{pmatrix} = \cos(\nu t)\,\sigma_x + \sin(\nu t)\,\sigma_y.$$

   Multiplying by $\frac{\hbar\Omega}{2}$ and adding $\frac{\hbar\nu_0}{2}\sigma_z$ gives (2.1).
2. Write $\psi_t = R\tilde{\psi}_t$. By *Linear Algebra*, Prop. 6.21, $\dot{R} = -\frac{\mathrm{i}\nu}{2}\sigma_z R$, so $\mathrm{i}\hbar\dot{R} = \frac{\hbar\nu}{2}\sigma_z R$. The product rule gives

$$\mathrm{i}\hbar\,\dot{\psi}_t - H(t)\psi_t = R\Big(\mathrm{i}\hbar\,\dot{\tilde{\psi}}_t + \tfrac{\hbar\nu}{2}\sigma_z\tilde{\psi}_t - R^\dagger H(t) R\,\tilde{\psi}_t\Big),$$

   where $\sigma_z R = R\sigma_z$ was used. By the first part, $R^\dagger H(t) R = \frac{\hbar}{2}(\nu_0\sigma_z + \Omega\sigma_x)$, and subtracting $\frac{\hbar\nu}{2}\sigma_z$ leaves $\tilde{H}$. So the bracket is $\mathrm{i}\hbar\dot{\tilde{\psi}}_t - \tilde{H}\tilde{\psi}_t$. The matrix R is invertible, so one side vanishes exactly when the other does.
3. Since $R(0) = \mathbb{1}$, the two functions start at the same vector ψ. By Lemma 1.11, the constant equation has exactly one solution, $\tilde{\psi}_t = e^{-\mathrm{i}t\tilde{H}/\hbar}\psi$. By the second part, the solutions of the two equations correspond one to one, so the driven equation also has exactly one, $R(t)\tilde{\psi}_t$. ∎

The vector $\tilde{\psi}_t$ describes the state in coordinates that turn with the transverse field. Its Bloch vector $\tilde{\mathbf{r}}$ is related to the Bloch vector of ψ_t by (2.2): $\mathbf{r} = \mathrm{Rot}_z(\nu t)\,\tilde{\mathbf{r}}$. In these coordinates the transverse field stands still along $\mathbf{e}_x$, and the term in σ_z has the coefficient Δ in place of ν_0. Without a drive ($\Omega = 0$) the Bloch vector precesses at the rate ν_0, and $\tilde{\mathbf{r}}$ at the rate $\Delta = \nu_0 - \nu$. The frame subtracts its own rate. A state at rest in the laboratory turns backwards, at the rate $-\nu$, in the frame.

The matrix $R(t)$ is diagonal, so $|\langle\downarrow|\psi_t\rangle| = \big|\langle\downarrow|\tilde{\psi}_t\rangle\big|$, and likewise for $|\uparrow\rangle$. The probabilities of a measurement along z are the same in both descriptions.

Starting at a time s instead of 0, the same argument gives $\psi_t = U(t,s)\psi_s$ with

$$U(t,s) = R(t)\, e^{-\mathrm{i}(t-s)\tilde{H}/\hbar}\, R(s)^\dagger,$$

the *evolution operator* from s to t. It is unitary, as a product of unitary matrices.

Self-check 2.3

1. With $\Omega = 0$, about which axis and at what rate does $\tilde{\mathbf{r}}$ precess?
2. Find $R(2\pi/\nu)$.

2.4 Rabi's formula

Theorem 2.2 (Rabi's formula). *Let $H(t)$ be the Hamiltonian (2.1), and put $\tilde{\Omega} = \sqrt{\Omega^2 + \Delta^2}$. For a spin prepared in the state $|\uparrow\rangle$ at $t = 0$, a measurement along z at the time t reads $|\downarrow\rangle$ with probability*

$$\mathrm{Prob}(\downarrow, t) = \frac{\Omega^2}{\tilde{\Omega}^2}\,\sin^2\Big(\frac{\tilde{\Omega}t}{2}\Big). \tag{2.5}$$

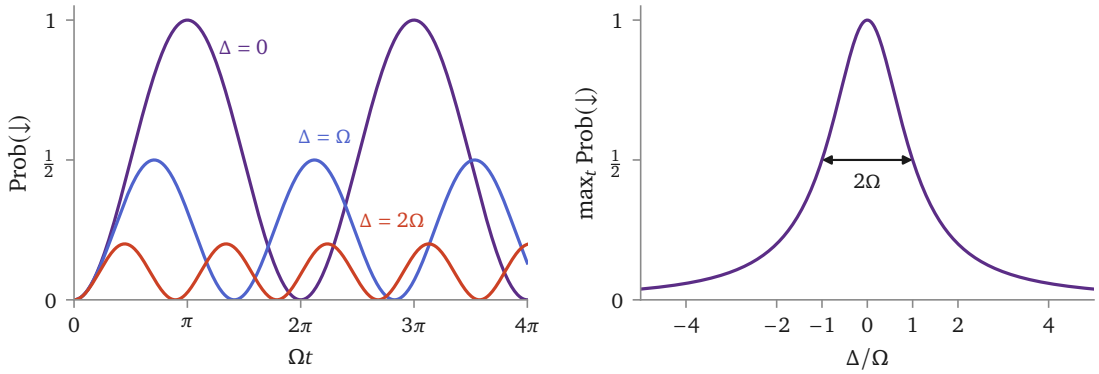


Figure 2.1: Rabi's formula. Left: $\mathrm{Prob}(\downarrow, t)$ against Ωt for $\Delta = 0$, Ω and 2Ω. Right: the largest value $\Omega^2/(\Omega^2 + \Delta^2)$ against Δ/Ω. It is $\frac{1}{2}$ at $\Delta = \pm\Omega$: the full width at half maximum is 2Ω.

Proof. Since $\Omega > 0$, also $\tilde{\Omega} > 0$, and (2.3) reads $\tilde{H} = \frac{\hbar\tilde{\Omega}}{2}\,\tilde{\mathbf{n}} \cdot \boldsymbol{\sigma}$ with the unit vector $\tilde{\mathbf{n}} = (\Omega, 0, \Delta)/\tilde{\Omega}$. By Lemma 2.1 and Theorem 1.12, the Bloch vector $\tilde{\mathbf{r}}$ turns about $\tilde{\mathbf{n}}$ at the rate $\tilde{\Omega}$, starting at $\mathbf{e}_z$. The formula of Theorem 1.12 with $\mathbf{r}_0 = \mathbf{e}_z$ has z-component

$$\tilde{r}_z(t) = \cos(\tilde{\Omega}t) + \bigl(1 - \cos(\tilde{\Omega}t)\bigr)\frac{\Delta^2}{\tilde{\Omega}^2},$$

since $\tilde{\mathbf{n}} \times \mathbf{e}_z$ has no z-component. The measurement along z has the same statistics in both descriptions, so by Corollary 1.8, using $1 - \Delta^2/\tilde{\Omega}^2 = \Omega^2/\tilde{\Omega}^2$,

$$\mathrm{Prob}(\downarrow, t) = \frac{1}{2}\bigl(1 - \tilde{r}_z(t)\bigr) = \frac{\Omega^2}{\tilde{\Omega}^2} \cdot \frac{1 - \cos(\tilde{\Omega}t)}{2} = \frac{\Omega^2}{\tilde{\Omega}^2}\,\sin^2\Bigl(\frac{\tilde{\Omega}t}{2}\Bigr).$$ ∎

Figure 2.1 draws the formula. The drive is *on resonance* when $\nu = \nu_0$, that is, $\Delta = 0$. Then $\tilde{\Omega} = \Omega$, and

$$\mathrm{Prob}(\downarrow, t) = \sin^2\Bigl(\frac{\Omega t}{2}\Bigr),$$

which reaches 1 at $t = \pi/\Omega$, for every $\Omega > 0$. In the rotating frame the static field is gone, and only the weak field is left. The vector $\tilde{\mathbf{r}}$ turns about $\mathbf{e}_x$, slowly, but all the way to $-\mathbf{e}_z$. This is why a weak field can turn a spin over.

Off resonance the probability never reaches 1. Its largest value, $\Omega^2/(\Omega^2 + \Delta^2)$, is $\frac{1}{2}$ when $|\Delta| = \Omega$. For a weak drive, $\Omega < \nu_0$, these are the two drive frequencies $\nu = \nu_0 \pm \Omega$, so the resonance has full width at half maximum 2Ω. The weaker the drive, the sharper the resonance, and the slower the turn. A static field is the case $\nu = 0$, with $\Delta = \nu_0$, and the formula gives the bound of § 2.1.

A resonant drive switched on for the time π/Ω is a *π-pulse*. It takes $|\uparrow\rangle$ to $|\downarrow\rangle$. In the frame, $e^{-\mathrm{i}\pi\sigma_x/2} = -\mathrm{i}\sigma_x$ and $-\mathrm{i}\sigma_x\,|\uparrow\rangle = -\mathrm{i}\,|\downarrow\rangle$. The diagonal matrix $R(t)$ then multiplies $|\downarrow\rangle$ by a phase only. Figure 2.2 draws its path in both descriptions. A drive for half that time, $\pi/(2\Omega)$, is a *$\pi/2$-pulse*. It turns $\tilde{\mathbf{r}}$ about $\mathbf{e}_x$ through a quarter turn, from $\mathbf{e}_z$ to $-\mathbf{e}_y$.

Two limits of the formula should be kept in mind. First, it is for a spin prepared up. From any other state, $\tilde{\mathbf{r}}$ still turns about $\tilde{\mathbf{n}}$ at the rate $\tilde{\Omega}$ (Theorem 1.12), but the probability is a different function of time. Second, a resonant drive with a fixed phase turns $\tilde{\mathbf{r}}$ about $\mathbf{e}_x$ only. From $\mathbf{e}_z$ it stays in the plane of $\mathbf{e}_y$ and $\mathbf{e}_z$, so one such pulse does not reach every state.

The phase of the drive removes the second limit. Suppose that the coils start the transverse field at the angle ξ, so that $\cos(\nu t)$ and $\sin(\nu t)$ in (2.1) become $\cos(\nu t + \xi)$ and $\sin(\nu t + \xi)$.

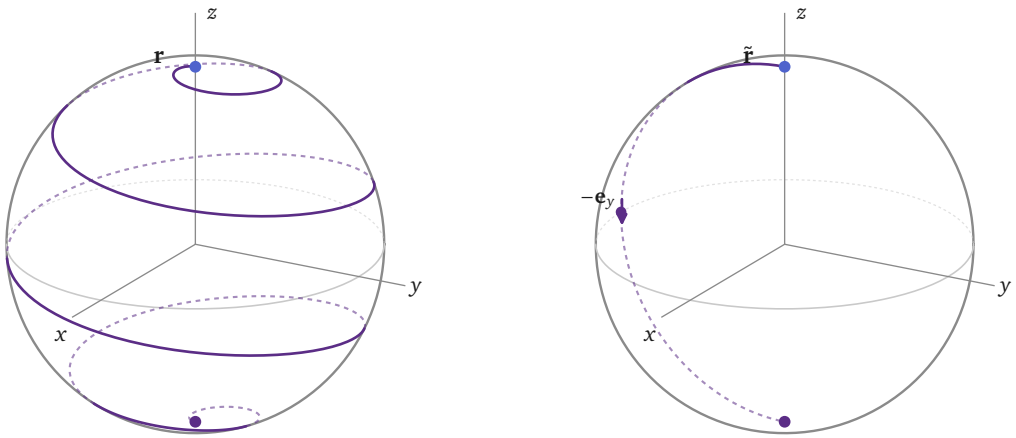


Figure 2.2: One π-pulse on resonance, drawn with $\nu = 8\Omega$. Left: the Bloch vector $\mathbf{r}$ in the laboratory spirals from the north pole to the south pole. Right: in the rotating frame, $\tilde{\mathbf{r}}$ turns about $\mathbf{e}_x$ along half a meridian, through $-\mathbf{e}_y$.

The proof of Lemma 2.1 goes through with $\cos\xi\,\sigma_x + \sin\xi\,\sigma_y$ in place of σ_x. On resonance, $\tilde{\mathbf{r}}$ then turns about the axis $(\cos\xi, \sin\xi, 0)$. A pulse that starts at $t = 0$ and lasts the time θ/Ω, with $0 \le \theta \le \pi$, takes it from $\mathbf{e}_z$ to

$$\tilde{\mathbf{r}} = \big(\sin\theta\sin\xi,\ -\sin\theta\cos\xi,\ \cos\theta\big),$$

the point with polar angle θ and azimuth $\xi - \pi/2$. Every point of the sphere has this form, so a pulse of the right duration and phase takes $\tilde{\mathbf{r}}$ from $\mathbf{e}_z$ to any point of the sphere.

Self-check 2.4

1. Let $\Delta = \sqrt{3}\,\Omega$. What is the largest probability of $|{\downarrow}\rangle$, and when is it first reached?
2. On resonance, what does a drive of duration $2\pi/\Omega$ do to the state $|{\uparrow}\rangle$?

2.5 A proton in an NMR magnet

Example 2.3 (A proton in an NMR magnet).

1. *The Hamiltonian.* A proton has a positive gyromagnetic ratio, $\gamma/2\pi = 42.577\ \text{MHz T}^{-1}$. By Lecture 1, its spin in a field $\mathbf{B}$ has $H = -\gamma\,\frac{\hbar}{2}\boldsymbol{\sigma}\cdot\mathbf{B}$. Take the field

$$\mathbf{B}(t) = -B_0\,\mathbf{e}_z - B_1\big(\cos(\nu t)\,\mathbf{e}_x + \sin(\nu t)\,\mathbf{e}_y\big), \qquad B_0 = 1\ \text{T}, \quad B_1 = 0.5\ \text{mT}.$$

   Then $H(t)$ is (2.1) with $\nu_0 = \gamma B_0$ and $\Omega = \gamma B_1$. The static field points along $-\mathbf{e}_z$, because for $\gamma > 0$ the Hamiltonian's axis is $-\mathbf{B}/|\mathbf{B}|$.
2. *The frequencies.* The symbol ν is an angular frequency, so a value in hertz is $\nu/2\pi$. Here $\nu_0/2\pi = 42.58$ MHz and $\Omega/2\pi = 21.29$ kHz, so $\Omega/\nu_0 = B_1/B_0 = 5\cdot 10^{-4}$.
3. *A static field.* The same 0.5 mT held still along $\mathbf{e}_x$ turns the spin over with probability at most $2.5\cdot 10^{-7}$ (§ 2.1).
4. *On resonance.* A π-pulse lasts $\pi/\Omega = 23.5\ \mu$s, and a $\pi/2$-pulse 11.7 μs.

5. *Off resonance.* A detuning $\Delta = \Omega$ moves $\nu/2\pi$ by 21.3 kHz, five parts in 10^4 of $\nu_0/2\pi$. Then $\tilde{\Omega} = \sqrt{2}\,\Omega$, and the probability of $|\downarrow\rangle$ reaches only $\frac{1}{2}$, first at $t = \pi/(\sqrt{2}\,\Omega) = 16.6\ \mu s$. At $\Delta = 10\,\Omega$ it never exceeds $\frac{1}{101} \approx 0.0099$.
6. *After a $\pi/2$-pulse.* On resonance, with the drive switched off at the end of the pulse, $\tilde{\mathbf{r}} = -\mathbf{e}_y$ stays put, since $\tilde{H} = 0$ when $\Delta = \Omega = 0$. By (2.2) the Bloch vector in the laboratory is

$$\mathbf{r}(t) = \mathrm{Rot}_z(\nu t)(-\mathbf{e}_y) = \big(\sin(\nu t),\ -\cos(\nu t),\ 0\big).$$

It turns in the xy-plane at the rate $\nu = \nu_0$.

The physics

Rabi, Zacharias, Millman and Kusch (1938) sent a beam of molecules through a strong magnetic field and a weak oscillating one of fixed radio frequency. When the strong field made the nuclear spins precess at that frequency, the molecules whose spins had turned over were lost from the detected beam. The known radio frequency and the field at the dip gave the nuclear magnetic moment. In papers published in 1946, Purcell, Torrey and Pound, and independently Bloch, Hansen and Packard, reported the same resonance in bulk matter: paraffin and water. In a water sample at room temperature that has reached thermal equilibrium in the static field, the protons have a small net magnetic moment along the field (Lecture 3). A resonant $\pi/2$-pulse turns this moment into the xy-plane. After the drive is switched off, the moment turns there and induces a voltage in a coil around the sample, at the frequency $\nu_0/2\pi$. That voltage is the signal of nuclear magnetic resonance (NMR), and of the magnetic resonance imaging (MRI) of protons in water.

Self-check 2.5

1. Which B_1 gives a proton a π-pulse of 10 μs?
2. With $B_1 = 0.5$ mT, how far may $\nu/2\pi$ be from $\nu_0/2\pi$ if the largest probability of $|\downarrow\rangle$ is to be at least $\frac{1}{2}$?

2.6 Ramsey's two pulses

Rabi's formula locates the frequency ν_0 to within the width 2Ω of the resonance. A sharper resonance needs a weaker drive, and so a longer pulse. In Ramsey's method, two short pulses turn the spin, and between them the spin precesses with the drive switched off, for a long time T. The sharpness then comes from T, not from Ω.

The oscillator that feeds the coils keeps running, and only its connection to the coils is switched. So during both pulses the field is the one of (2.1), as a function of the same time t, and one rotating frame $R(t)$ serves throughout. The proof of Lemma 2.1 does not use $\Omega > 0$. So while the drive is off, the frame Hamiltonian is $\tilde{H} = \frac{\hbar\Delta}{2}\sigma_z$.

Proposition 2.4 (Ramsey's fringes)**.** *A spin prepared in the state $|\uparrow\rangle$ receives a $\pi/2$-pulse, evolves with the drive off for the time T, and receives a second $\pi/2$-pulse, the drive keeping its phase. Suppose that each pulse turns $\tilde{\mathbf{r}}$ about $\mathbf{e}_x$ through exactly $\pi/2$. Then a measurement along z at the end reads $|\downarrow\rangle$ with probability*

$$\mathrm{Prob}(\downarrow) = \cos^2\Big(\frac{\Delta T}{2}\Big). \tag{2.6}$$

Proof. We follow $\tilde{\mathbf{r}}$ through the three steps.

1. The first pulse takes $\mathbf{e}_z$ to $-\mathbf{e}_y$, as in § 2.4.
2. During the wait, by Theorem 1.12 with the axis $\mathbf{e}_z$ if $\Delta > 0$ and $-\mathbf{e}_z$ if $\Delta < 0$, $\tilde{\mathbf{r}}$ turns about $\mathbf{e}_z$ through the angle ΔT, to $\mathrm{Rot}_z(\Delta T)(-\mathbf{e}_y) = (\sin\Delta T,\ -\cos\Delta T,\ 0)$.
3. A quarter turn about $\mathbf{e}_x$ takes $(x, y, 0)$ to $(x, 0, y)$. So after the second pulse, $\tilde{\mathbf{r}} = (\sin\Delta T,\ 0,\ -\cos\Delta T)$.

The measurement along z has the same statistics in both descriptions (§ 2.3). By Corollary 1.8,

$$\mathrm{Prob}(\downarrow) = \tfrac{1}{2}\big(1 - \tilde{r}_z\big) = \tfrac{1}{2}\big(1 + \cos(\Delta T)\big) = \cos^2\Big(\frac{\Delta T}{2}\Big). \qquad \blacksquare$$

The ideal pulses are an approximation, good when $|\Delta| \ll \Omega$. Then the axis of $\frac{\hbar}{2}(\Delta\sigma_z + \Omega\sigma_x)$ is close to $\mathbf{e}_x$, and each pulse is close to a quarter turn about it.

As the drive frequency varies, the probability (2.6) rises and falls in a row of *fringes*. It is 1 when ΔT is a multiple of 2π, and 0 when ΔT is an odd multiple of π. The central fringe falls to half its height at $|\Delta| = \pi/(2T)$, so its full width at half maximum is π/T in Δ. The width is set by the wait T alone, and not by the strength of the drive. All the fringes have the same height, so (2.6) alone does not say which one is the central fringe.

Why does the wait matter, if nothing is driven during it? Throughout the wait $\tilde{r}_z = 0$, and a reading along z would give ± 1 with probability $\frac{1}{2}$, whatever Δ is. The detuning shows only in the azimuth of $\tilde{\mathbf{r}}$, the relative phase of $|\uparrow\rangle$ and $|\downarrow\rangle$ (Example 1.13). The second pulse turns the angle gained during the wait into a height, and the reading along z sees it.

The physics

Ramsey proposed the method in 1949 for beams of molecules. The two pulses were two short regions of oscillating field at the two ends of the apparatus, and the wait was the molecules' flight between them. Molecules of different speeds spend different times T in flight, so their fringes have different spacings. The average damps the side fringes, while the central one, at $\Delta = 0$, keeps its full height for every flight time.

Self-check 2.6

1. For a wait T, what is the smallest $|\Delta| > 0$ at which $\mathrm{Prob}(\downarrow) = 0$?
2. On resonance, the second pulse has the phase $\xi = \pi$, so that it turns $\tilde{\mathbf{r}}$ about $-\mathbf{e}_x$. What is $\mathrm{Prob}(\downarrow)$ at the end?

2.7 Circular and linear drives

Remark 2.5 (A linear drive)**.** A single coil along x makes a field that oscillates along a line. For the proton of Example 2.3 take $-2B_1\cos(\nu t)\,\mathbf{e}_x$. It adds $\hbar\Omega\cos(\nu t)\,\sigma_x$ to the Hamiltonian, the sum of two circular terms that turn in opposite senses:

$$2\cos(\nu t)\,\sigma_x = \big(\cos(\nu t)\,\sigma_x + \sin(\nu t)\,\sigma_y\big) + \big(\cos(\nu t)\,\sigma_x - \sin(\nu t)\,\sigma_y\big).$$

The rotating frame removes the time dependence of the first one only. In the frame, the second becomes $\frac{\hbar\Omega}{2}\big(\cos(2\nu t)\,\sigma_x - \sin(2\nu t)\,\sigma_y\big)$ and turns at the rate 2ν. Lemma 2.1 and Rabi's formula are exact for a circular drive only. Dropping the second term is the *rotating-wave approximation*. It is an approximation, and how good it is depends on Ω/ν and on the time. For example, a numerical solution of the Schrödinger equation with $\Omega/\nu_0 = 0.01$ and $\nu = \nu_0$ turns a spin prepared up over with probability above 0.999 at the end of a pulse of duration

π/Ω. The small shift of the resonance that the second term causes is computed on Exercise Sheet 2.

Self-check 2.7

1. In the rotating frame, at what rate does the second circular part of a linear drive turn, and in which sense about $\mathbf{e}_z$?

The driven qubit, as a quantum system

States and observables: those of Lecture 1, the points of the Bloch sphere and the self-adjoint matrices $a\mathbb{1} + b\,\mathbf{m}\cdot\boldsymbol{\sigma}$. *Evolution:* $\mathrm{i}\hbar\dot{\psi}_t = H(t)\psi_t$. Under the circular drive (2.1), $\psi_t = R(t)\,e^{-\mathrm{i}t\tilde{H}/\hbar}\psi_0$, and in the rotating frame the sphere turns about $(\Omega, 0, \Delta)/\tilde{\Omega}$ at the rate $\tilde{\Omega}$ (Lemma 2.1). *Measurement:* as in Lecture 1. From $|\uparrow\rangle$, the reading $|\downarrow\rangle$ has probability $(\Omega/\tilde{\Omega})^2 \sin^2(\tilde{\Omega}t/2)$ (Theorem 2.2).

Summary.

- *The rotating frame*: the circular drive becomes a constant Hamiltonian, so for each initial vector the Schrödinger equation has exactly one solution, and it is known (Lemma 2.1).
- *Rabi's formula*: under a circular drive with any $\Omega > 0$, a spin prepared up is turned over on resonance at the time π/Ω. Off resonance the probability of the turn stays below 1 (Theorem 2.2).
- *Ramsey's fringes*: from spin up, two ideal quarter turns about $\mathbf{e}_x$, a wait T apart with the drive's phase kept, turn the spin over with probability $\cos^2(\Delta T/2)$ (Proposition 2.4).

Lecture 3 asks what happens when the preparation is not known exactly, as for the atoms straight from the oven.

Self-check answers.

2.1 1. $0.01/1.01 \approx 0.0099$. 2. The two at $\mathbf{r} = \pm\mathbf{n}$, with $\mathbf{n}$ along $(\Omega, 0, \nu_0)$.

2.2 1. $\frac{\hbar}{2}(\nu_0\sigma_z + \Omega\sigma_y)$. 2. The proof of Lemma 1.11 needs H self-adjoint only at the time of differentiation.

2.3 1. About $\mathbf{e}_z$ at the rate $\Delta = \nu_0 - \nu$. 2. $e^{-\mathrm{i}\pi\sigma_z} = -\mathbb{1}$.

2.4 1. Here $\tilde{\Omega} = 2\Omega$, so the largest probability is $\frac{1}{4}$, first at $t = \pi/(2\Omega)$. 2. It returns the state to $|\uparrow\rangle$: $e^{-\mathrm{i}\pi\sigma_x} = -\mathbb{1}$, a phase.

2.5 1. $\Omega = \pi/(10\ \mu\mathrm{s})$, so $B_1 = \Omega/\gamma \approx 1.17$ mT. 2. By at most $\Omega/2\pi \approx 21.29$ kHz: the condition is $|\Delta| \le \Omega$.

2.6 1. $|\Delta| = \pi/T$, where $\Delta T = \pm\pi$. 2. 0: the second pulse undoes the first, and the spin is back at $|\uparrow\rangle$.

2.7 1. At the rate 2ν, clockwise seen from the tip of $\mathbf{e}_z$.

References

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